WEEK 10- CUSTOMER CHURN PREDICTION

Problem Introduction

Customer churn is a critical operational metric for subscription-based businesses, representing the rate at which customers discontinue their services. Retaining an existing customer is significantly more cost-effective than acquiring a new one. This project analyzes a 1-million-record customer dataset to identify behavioral and demographic indicators of churn and builds predictive models to proactively identify at-risk customers. The ultimate goal is to convert predictive insights into targeted retention strategies.

Concepts Applied

- **Exploratory Data Analysis (EDA):** Identifying underlying distributions, correlations, and target imbalances.
- **Data Preprocessing:** Label encoding for categorical text variables, median imputation for missing values, and Standard Scaling to normalize feature magnitudes.
- **Class Imbalance Resolution:** Applying SMOTE (Synthetic Minority Over-sampling Technique) to ensure the models do not become biased toward the majority "no-churn" class.
- **Supervised Machine Learning:** Utilizing Logistic Regression for baseline probabilistic classification and Decision Trees for capturing non-linear relationships.
- **Performance Metrics:** Moving beyond simple accuracy by leveraging Precision, Recall, F1-Score, and AUC-ROC to evaluate model effectiveness on imbalanced data.

Data Preprocessing Steps

1. **Handling Missing Data:** Missing numerical values were imputed using the median to avoid outlier skew, while missing categorical values were filled with the mode.
2. **Encoding:** Binary categories (e.g., Gender, Partner) and multi-class categories (e.g., Contract Type, Internet Service) were transformed into numerical arrays using LabelEncoder.
3. **Scaling:** Features like Tenure and MonthlyCharges were standardized using StandardScaler to have a mean of 0 and variance of 1, preventing high-magnitude variables from dominating the Logistic Regression cost function.

4. **Balancing:** The synthetic oversampling of churned customers ensured the models could learn the minority class boundaries effectively.

Model Implementation & Evaluation

Both models were trained using 5-fold cross-validation on the balanced training set and evaluated on the holdout test set.

- **Logistic Regression:** Provided a strong baseline with high interpretability. While it achieved reliable overall accuracy, its **Recall** for the churn class was heavily scrutinized, as failing to identify a churning customer (False Negative) is a highly penalized error in this business context.
- **Decision Tree:** Captured complex interactions between variables like contract length and monthly charges. By restricting max_depth, the model avoided overfitting the 1-million-row dataset.
- **Trade-off Analysis:** In telecom churn, **Recall** is generally prioritized over **Precision**. A false positive (offering a retention discount to a happy customer) costs less than a false negative (losing a high-value customer entirely). The ROC curve demonstrated strong discriminatory ability, with the Decision Tree slightly outperforming Logistic Regression in isolated feature splits.

Key Insights and Actionable Recommendations

Key Insights:

1. **Contract Type is the Strongest Predictor:** Customers on month-to-month contracts churn at drastically higher rates than those on one- or two-year commitments.
2. **Tenure & Churn Inversely Correlate:** The first 6 months of the customer lifecycle are the most volatile. If a customer passes the one-year mark, their probability of churning drops exponentially.
3. **High Monthly Charges Drive Attrition:** Customers with premium add-ons but no long-term contract form the highest-risk segment.

Business Strategies to Reduce Churn:

1. **Incentivize Long-Term Contracts:** Offer temporary discounts, free streaming add-ons, or priority support to month-to-month users who agree to upgrade to a 1-year contract.
2. **Early Intervention Program:** Deploy proactive customer success check-ins at Month 3 and Month 6. Ensure onboarding is smooth and customers are actually utilizing the services they pay for.

3. **Dynamic Pricing & Alerts:** Use the predictive model to score active customers weekly. If a customer crosses a 70% churn probability threshold (often triggered by high bills and low tenure), automatically generate a targeted retention email offering a tailored loyalty discount.